ANSWER KEY — Integration Lesson 38 Test: Growth, Decay & Logistic

For the grader · full working shown · give credit for correct method with small arithmetic slips

1. A colony of yeast cells, N, grows at a rate proportional to the number of cells present. Write the differential equation and its solution in terms of N_0.

> **Work:** "rate" → dN/dt; "proportional to the number present" → = kN. So `dN/dt = kN` with k > 0. Solution (Lesson 37): $N = Ce^{kt}$, and at t = 0, $N(0) = C \cdot e^0 = C$, so $C = N_0$. **Answer: dN/dt = kN (k > 0), with solution $N = N_0e^{kt}$. Grader note:** accept dN/dt = kN without stating k > 0; the constant must be the starting amount, not an unnamed C.

2. A loaf of bread at temperature T sits in a 15° kitchen and cools at a rate proportional to how much hotter it is than the kitchen. Write the DE and explain the minus sign.

> **Work:** the gap above room temperature is T − 15. Newton's Law of Cooling: `dT/dt = −k(T − 15)`, k > 0. Sign check: at T = 180 the right side is −165k < 0, so the bread is cooling ✓; if T were 10 (colder than the kitchen) the right side is +5k > 0, warming toward 15 ✓. **Answer: dT/dt = −k(T − 15), k > 0 — the minus sign makes a positive gap produce a falling temperature.**

3. 250 lily pads, doubling every 5 days. (a) Formula for P(t) in base 2. (b) Pads after 20 days?

> **Work:** (a) doubling every d = 5 days → `P = 250·2^(t/5)` (the exponent t/5 counts doublings). (b) t = 20 → 20/5 = 4 doublings → $P = 250 \cdot 2^4 = 250 \cdot 16 = 4000$. Ladder check: 250 → 500 (5 d) → 1000 (10 d) → 2000 (15 d) → 4000 (20 d) ✓. **Answer: (a) P = 250·2^(t/5)   (b) 4000 lily pads.**

4. Medicine with an 8-hour half-life; 640 mg dose. How much remains after 24 hours?

> **Work:** number of half-lives = 24/8 = 3. Remaining = $640 \cdot (1/2)^3 = 640/8 = 80$. Ladder check: 640 → 320 (8 h) → 160 (16 h) → 80 (24 h) ✓. **Answer: 80 mg. Grader note:** a common slip is dividing by 3 (640/3) instead of halving three times.

5. A culture grows exponentially from 400 to 3200 cells in 12 hours. Find the doubling time and check it.

> **Work:** growth factor = 3200/400 = 8 = 2^3, so 3 doublings happened in 12 hours → doubling time = 12/3 = 4 hours. Model: `P = 400·2^(t/4)`. Check by substituting t = 12: $400 \cdot 2^{12/4} = 400 \cdot 2^3 = 400 \cdot 8 = 3200$ ✓. (Base-e version if wanted: k = ln 2/4 ≈ 0.173 per hour.) **Answer: doubling time = 4 hours.**

6. `P = 60·2^(t/4)` (t in hours). When does P reach 1920?

> **Work:** isolate the exponential FIRST: 60·2^(t/4) = 1920 → 2^(t/4) = 32. Recognize $32 = 2^5$ → t/4 = 5 → t = 20. (ln route: (t/4)·ln 2 = ln 32 = 5 ln 2 → t/4 = 5 → t = 20, same.) Check: $60 \cdot 2^{20/4}$ = 60·32 = 1920 ✓. **Answer: t = 20 hours. Grader note:** taking ln before dividing by 60 is messy but not wrong; penalize only a wrong final answer.

7. (a) Rewrite N = 250·2^(t/5) as $N = 250 \cdot e^{kt}$. (b) For `y = 80·e^(−(ln 2/7)t)` (t in days), find the half-life and the amount left after 21 days.

> **Work:** (a) since $2 = e^{\ln 2}$, 2^(t/5) = $(e^{\ln 2})$^(t/5) = e^((ln 2/5)t), so k = ln 2/5 ≈ 0.693/5 ≈ 0.1386 per day. Check: N(5) = $250e^{\ln 2}$ = 250·2 = 500 — doubled in 5 days ✓.
> (b) Decay with k = −ln 2/h means h = half-life, so h = 7 days (rule: half-life = ln 2 ÷ |k| = ln 2 ÷ (ln 2/7) = 7). Then 21 days = 3 half-lives → y = $80 \cdot (1/2)^3$ = 10. Check with the exponential: 80·e^(−(ln 2/7)·21) = 80·e^(−3 ln 2) = $80 \cdot 2^{-3}$ = 80/8 = 10 ✓. **Answer: (a) k = ln 2/5 ≈ 0.1386 (b) half-life 7 days; 10 grams remain.**

8. Roast at 180° in a 20° kitchen; 100° after 12 minutes. (a) Half-life of the gap. (b) Temperature after 36 minutes. (c) Will it ever be exactly 20°?

> **Work:** (a) starting gap = 180 − 20 = 160°. After 12 min the gap is 100 − 20 = 80° — exactly half of 160. So the GAP's half-life is 12 minutes (i.e. e^(−12k) = 1/2).
> (b) 36 min = 36/12 = 3 gap-halvings: 160 → 80 → 40 → 20. Temperature = room + gap = 20 + 20 = 40°. Formula check: $T = 20 + 160 \cdot (1/2)^{36/12}$ = 20 + 160/8 = 20 + 20 = 40 ✓.
> (c) T = 20 exactly would need 160e^(−kt) = 0, i.e. e^(−kt) = 0 — impossible, since e raised to any power is positive. So T > 20 for every finite t, but lim T = 20 + 160·0 = 20 as t → ∞: the roast approaches 20° as a horizontal asymptote. **Answer: (a) 12 minutes (b) 40° (c) No — it approaches 20° but never reaches it.**
> **Grader note (classic error):** students apply the half-life to the TEMPERATURE instead of the gap: 180 → 90 (12 min) → 45 → 22.5. That is wrong, and absurd — continued, it drops below room temperature toward 0°. The half-life belongs to u = T − 20 only; add the 20 back at the very end.

9. `dP/dt = 0.3P(1 − P/900)`, P(0) = 150. (a) Carrying capacity? (b) lim P? (c) Fastest-growth population and dP/dt there? (d) dP/dt at P = 300?

> **Work:** match to $kP(1 - P/L)$: $k = 0.3$, $L = 900$.
> (a) Carrying capacity $L = 900$ rabbits. (At $P = 900$ the brake is $1 - 1 = 0$, so growth stops exactly there.)
> (b) P starts at 150, below the ceiling, so it climbs the S-curve: $\lim P = 900$ rabbits as $t \to \infty$.
> (c) Fastest growth at $P = L/2 = 450$ (the parabola $kP - kP^2/L$ peaks midway between its roots 0 and L). There $dP/dt = 0.3 \cdot 450 \cdot (1 - 450/900) = 135 \cdot (1/2) = 67.5$ rabbits per year.
> (d) $dP/dt = 0.3 \cdot 300 \cdot (1 - 300/900) = 90 \cdot (2/3) = 60$ rabbits per year — less than 67.5, as it must be, since 450 is the peak ✓. **Answer: (a) 900 (b) 900 (c) P = 450, dP/dt = 67.5 rabbits/year (d) 60 rabbits/year.**
> **Grader note (classic error):** do not accept 900 for part (c) — the carrying capacity is where growth STOPS; half of it is where growth is fastest. Mixing up L and L/2 is the single most common logistic mistake.

10. Valley supports at most 4000 elk, logistic with k = 0.25/year. (a) DE. (b) Fastest herd size and rate. (c) Find the student's error. (d) 5000 elk released — long run?

> **Work:** (a) "can support at most 4000" is the carrying capacity: `dP/dt = 0.25P(1 − P/4000)`.
> (b) Fastest at $P = L/2 = 2000$ elk; there $dP/dt = 0.25 \cdot 2000 \cdot (1 - 2000/4000) = 500 \cdot (1/2) = 250$ elk per year. (Symmetry check: at $P = 1000$, $0.25 \cdot 1000 \cdot (3/4) = 187.5$, and at $P = 3000$, $0.25 \cdot 3000 \cdot (1/4) = 187.5$ — equal on both sides of the peak ✓.)
> (c) The error: more elk does NOT mean faster growth, because the brake factor $(1 - P/4000)$ shrinks as P rises. At $P = 4000$ the brake is exactly 0, so $dP/dt = 0.25 \cdot 4000 \cdot 0 = 0$ — growth is at its SLOWEST (stopped), not fastest. The peak is at $P = 2000$.
> (d) At $P = 5000$ the brake is $1 - 5000/4000 = -0.25$, so $dP/dt = 0.25 \cdot 5000 \cdot (-0.25) = -312.5$ elk/year: the herd shrinks. It keeps shrinking while $P > 4000$ and levels off at the carrying capacity, so $P \to 4000$. **Answer: (a) dP/dt = 0.25P(1 − P/4000) (b) P = 2000, 250 elk/year (c) at P = 4000 the brake is 0, so growth stops; fastest is at L/2 = 2000 (d) the herd declines toward 4000 elk.**
> **Grader note:** in (d) full credit requires the limit 4000 (not 0 and not "keeps falling forever"); the population approaches the carrying capacity from above.